# Gaurika Mahajan

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## Education

**University of Toronto + PEY Co-op** Toronto, ON
BASc in Computer Engineering, Minor in Artificial Intelligence *Sept 2022 – May 2027*
*Relevant Courses: Computer Networks, Data Structures and Algorithms, Databases, Operating Systems, Deep Learning*

## Technical Skills

**Languages**: Java, Python, Go, C++, Javascript, HTML, CSS, C, MATLAB
**Tools**: Jenkins, Docker, Kubernetes, MongoDB, PostgreSQL, Splunk, Datadog, SonarQube
**Frameworks**: React, Express, Node.js
**Cloud**: AWS (S3, EC2, Lambda, Redshift, DynamoDB)
**Dev Practices**: Git, CI/CD, Testing, OOP, Design Patterns, Agile, UML, Linux/Unix, Windows, Prompt Engineering

## Work Experience

**Software Engineering Intern** May 2025 – Present
Zynga Toronto, ON

- Migrated critical heartbeat API from legacy PHP endpoints to modern event-driven architecture, improving system reliability **by 40%** and reducing API response latency while maintaining backward compatibility for **10+ game titles**
- Built centralized CI/CD infrastructure by integrating SonarQube across **8 repositories** and creating reusable Docker images deployed on EKS, reducing code duplication **by 65%** and standardizing deployment pipelines for backend services
- Led cross-team system migration to replace legacy RPM distribution infrastructure with DNF-based solution, eliminating single point of failure and improving package deployment success rate from **70% to 98%**

**Full Stack Web Developer** May 2024 – Aug 2024
UofT Engineering Orientation Website Toronto, ON

- Developed and launched **React/Redux** web application serving **1,000+ engineering students** using SDLC best practices, building **Express REST API** with real-time leaderboard functionality and data visualizations
- Architected scalable backend infrastructure using MongoDB with optimized indexing **(40% query performance improvement)** and **Redis** caching for background subscriptions, enhancing system scalability **by 20%**
- Integrated reliable payment flows via **Stripe API** and webhooks to process **$150,000+** in transactions

**Compliance Intern** May 2024 – Aug 2024
Fresenius Kabi USA Chicago, IL

- Designed monorepo architecture to consolidate data across 7 repositories into a single pipelined workflow, **improving data retrieval speed by 95%**; automated alerts for call-to-action (CTA) responses, enhancing operational stability
- Developed data visualization pipeline, in **SmartSheet** and **SharePoint**, used in quarterly reports that speeds up deadline tracking, trend identification, and vendor/product reporting, **by 80%**
- Analyzed **unstructured data** (100k+ datapoints across 12 years) to create database schema, track quality agreements, and extract insights that improved product launch forecasting **from 2 days to seconds**

## Leadership

**Software Developer Director** April 2024 – Present
UofT You're Next Career Network

- Lead team of 10 developers in expanding **React/Firebase** platform serving 3,500+ students to manage event sign-ups
- Defined product roadmap, created technical specs for feature development, and designed user-centric interface in Figma

**Web Development Director** Aug 2023 – Aug 2024
UofT Women in Science & Engineering

- Led website development and infrastructure improvements, enhancing user experience for **500+ members**
- Architected automated email management system using Gmail APIs, reducing manual administrative overhead **by 60%**

## Projects

**Skin Cancer Classifer** | Python, PyTorch, TensorFlow, Google Colab, GitHub

- Developed a AI-driven CNN-based skin cancer classifier, achieving **80% test accuracy** on dermatoscopic images
- Integrated unsupervised learning techniques for data augmentation and feature extraction to enhance training data quality
- Led data collection, model creation, training, and hyperparameter optimization for improved classification performance
- Produced detailed reports showcasing the model's architecture, performance metrics, & actionable insights

**GIS Mapping Software** | C++, Git, Valgrind

- Built and optimized a GIS application in a team of 3 using an **Agile-inspired approach** and **OpenStreetMap API**
- Implemented **A* and Dijkstra algorithms** for efficient pathfinding, leveraging **multi-threading** to cut latency by **200%**